

Viscous fluid flow around a circular cylinder

Objectives

Demonstrate characteristics of viscous fluid flow around a circular cylinder (e.g., pressure profiles and boundary layer flows). Evaluate the drag force exerted on a cylinder.

Description

This experiment uses an educational wind tunnel of the open-circuit type. There is a force balance that can be used to measure aerodynamic forces acting on objects placed in the test section. Provisions have also been made to allow measurement of pressures at various points on objects placed in the flow (though these objects must have pressure taps located on them).

Background

Read a fluid dynamics text about flows over cylinders. Topics of interest are surface pressure profiles, laminar and turbulent boundary layers, Reynolds numbers, drag coefficients, and boundary layer separation. You should also read about dimensionless numbers and scaling.

Safety Guidelines

Compliance with safety instructions is mandatory. General safety information is listed below. There is also safety information at various points in this handout. Read all of this information carefully as well as any safety guidelines posted in the laboratory.

- Teaching Assistants for EME 109 must be present when any of the equipment or experiments are operated.
- Doorways must be kept clear.
- Power tools cannot be used when you are alone in the laboratory.
- Electrical wiring changes cannot be done while you are alone in the laboratory.
- If you are supplied eye or ear protection, you must use it.
- Safety goggles or glasses must be worn when using power tools or chemicals.
- Ventilation fans must be on when using any chemicals or operating any combustion engines in the laboratory.
- A carbon monoxide (CO) monitor must be used when operating any combustion engines in the laboratory.

- All students (and Teaching Assistants) must be familiar with the location and operating instructions for the nearest fire extinguisher.
- Full length pants (or equivalent) and closed toe/heel shoes must be worn by all people who occupy or enter the laboratory. The area of skin between the pants and the shoes must not be exposed.

Warning: There should be no need to unplug or plug in the power cords once the laboratory sessions have started. Only the TA's should handle the power cords.

Warning: Only the TA should operate controls not located on the main control panel.

Warning: Stay away from the fan area while the fan motor is operating.

Warning: Do not unlatch or open the wind tunnel doors while the fan is in motion.

Warning: Always wear safety glasses and hearing protection when operating the laboratory equipment!

Warning: Always make sure that all loose objects are removed from the interior of the wind tunnel before turning on the fan.

Warning: Never allow the airspeed to exceed the value selected for this experiment. The drag forces on the cylinder may be too large for airspeeds higher than this. Always monitor the manometer reading and make adjustments as necessary to keep the airspeed at the required value.

Experimental Procedure

1. Put on your safety glasses.
2. Record the atmospheric pressure and temperature on your data sheet.
3. You will perform experiments with the test section airspeed specified by your TA. For this airspeed, calculate the reading that will apply to the manometer (use the Bernoulli equation for the pitot-static probe). Record these values below (include units).

Airspeed _____

Airspeed Manometer Reading _____

4. If not already in place on the cylinder, cut a short length of string (≈ 2 cm). Firmly tape the string onto the cylinder at the same angle as the pressure tap (do not cover the pressure tap). You may also tape string at other locations. The purpose of the string is to allow the flow direction to be visualized.
5. **IMPORTANT:** Make sure all loose objects have been removed from the wind tunnel.
6. Put on your hearing protection.
7. Zero the drag measurement system.

8. Check to make sure that the fan speed control knob is rotated counterclockwise as far as possible.
9. Start the fan motor.
10. Slowly increase the airspeed until the manometer reaches the appropriate value. Rotate the fan control knob clockwise to increase airspeed and counterclockwise to decrease airspeed.
11. Let the fan run for about 5 minutes to allow the motor control system to stabilize. Make adjustments as necessary to keep the airspeed at the required value.
12. Slowly rotate the cylinder so the pressure tap is at angles of 0, , 10, 20, ..., 180 degrees. Ask your TA to show you the correct way to rotate the cylinder. At each angle, record the manometer values for the cylinder surface pressure and the airspeed. Also record the drag and sketch the string behaviors.
13. The pitot-static probe performance is sensitive to the orientation of the probe with respect to the flow (ideally, it should be aligned with the flow). Investigate the pitot-static probe performance by rotating it to several different angles. At each angle, record the manometer reading.
14. Evaluate the variations in airspeed along the wind tunnel test section. Do this by performing a pitot-static probe traverse. Starting with the pitot-static probe near the test section top wall, move it vertically across the test section. Record the manometer reading about every 2 cm. Note any variations in the cylinder string behavior as the pitot-static probe changes positions.
15. Reduce the airspeed to zero by turning the fan control knob counterclockwise.
16. Stop the fan motor.
17. Return the pitot-static probe to its original location.

Temperature _____

Pressure _____

Humidity _____

Cylinder Data

Pressure Tap Angle	Cylinder Manometer	Airspeed Manometer	Measured Drag

Pitot-Static Probe Angle Data

Probe Angle	Manometer Reading

Pitot-Static Probe Traverse Data

Probe Location	Manometer Reading

Observations

Sketches